

Un análisis de corpus sobre el marcado diferencial de objeto en hablantes de L2 español con L1 alemán

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■ **DOM:** The use of the marker *a* with direct objects

- (1) a. Encontré un problema.
found a problem.DO
'I found a problem.'
- b. Encontré *a* un superviviente.
found DOM a survivor.DO
'I found a survivor.'

(Fábregas 2013: 2)

- (2) a. (Te) vi *a* ti.
(you.DO) saw DOM you.IDO
'I saw you.'
- b. Te vi **(a)* ti.
(you.DO) saw **(DOM)* you.IDO

(Fábregas 2013: 30)

- DOM is related to case-marking, but not a phenomenon of case-marking
- no obligatory *a*-marking of all direct objects (Fábregas 2013: 30)

- Dependent upon specific syntactic, semantic, and pragmatic factors
 - animacy
 - specificity and definiteness
 - topicality
 - etc.

- See Kabatek, Obrist, and Wall 2021 among others for an overview

Animacy

- (3) a. Marcelo vio *a* Mónica.
Marcelo saw DOM Mónica.DO. [+animate]
'Marcelo saw Mónica.'
- b. La crisis afectó la moral.
the crisis affected the moral.DO. [-animate]
'The crisis affected the moral.'
- c. La bomba destruyó una iglesia.
the bomb destroyed a church.DO. [-animate]
'The bomb destroyed a church.'

(García García 2005: 23)

Specificity

- (4) a. El adjetivo modifica *al* sustantivo.
the adjective modifies DOM-the noun.DO. [-animate, +specific]
'The adjective modifies the noun.'
- b. Cosecharon *al* maíz.
harvested DOM-the maize.DO. [-animate, +specific]
'They harvested the maize.'

(García García 2005: 64; Pomino 2012: 309)

- Interaction between semantic scales of animacy and specificity (animacy as mostly dominant), see Referentiality scale presented in Heusinger 2008

personal pronoun > proper noun > definite NP > non-specific NP

- Topicality as a possible explanation of the optionality of DOM
- a-marked DOs have properties of discourse topics
- *a* as a possible topicality marker

- (5) a. *A Pepe vemos.*
DOM Pepe.DO see
'We see Pepe.'
- b. *A Pepe le ayudamos.*
DOM Pepe.DO.TOP him help
'We help Pepe.'

(Leonetti 2004: 87)

- *a*-marking is obligatory in cases of clitic left dislocation, clitic doubling, and fronted objects

- Transitivity-related properties of the verb as another factor of DOM cf. García García 2014, Kabatek, Obrist, and Wall 2021, Heredero 2022

Aspect (Telicity)

- (6) a. María saludó $a/*\emptyset$ un compañero.
María greeted.[+TEL] DOM/ $\emptyset$ a colleague.DO
'María greeted a colleague.'
- b. Observaron $a/\emptyset$ un niño.
watched.[-TEL] DOM/ $\emptyset$ a child.DO
'They watched a child.'

(Torrego 1999: 1789)

Affectedness

- Tsunoda (1985):
 - **action** > **perception** > **pursuit** > **knowledge** > **feeling**
- Hopper and Thompson (1980):
 - **effect on patient** > **perception** > **pursuit** > **knowledge** > **feeling**
- the degree of "force-transmission from the agent to the patient" evoked by the verb (Kabatek, Obrist, & Wall 2021: 149)

direct effect on patient		perception		pursuit		knowledge		feeling	
<i>matar</i>	'kill'	<i>mirar</i>	'watch'	<i>buscar</i>	'search'	<i>conocer</i>	'know'	<i>querer</i>	'want'
<i>herir</i>	'hurt'	<i>oír</i>	'hear'	<i>seguir</i>	'follow'	<i>entender</i>	'understand'	<i>temer</i>	'fear'
<i>golpear</i>	'hit'	<i>escuchar</i>	'listen'	<i>esperar</i>	'await'	<i>aprender</i>	'learn'	<i>odiar</i>	'hate'

Table: Examples of verbs along the Transitivity Scale, based on Wall and Obrist (2021)

→ DOM as a potential marker of transitivity

- Germanic languages generally lack DOM
- polyfunctionality of the marker *a* in Spanish
 - marker for indirect objects
 - appears in lexicalised and grammaticalised forms
 - locative preposition

- (7) a. Juan le da un libro a Ana.
Juan her gives a book.DO to Ana
Juan gives Ana a book.
- b. Juan y Ana van a un concierto.
Juan and Ana go to.PREP.LOC a concert
'Juan and Ana are going to a concert.'
- c. Juan y Ana van a ir a un concierto.
Juan and Ana go to.PREP.FUT go to.PREP.LOC a concert
'Juan and Ana are going to go to a concert.'

(Ciovarnache & Avram 2013)

- low salience of the marker *a*

→ High vulnerability to L1-transfer effects and overall acquisitional difficulties

Ticio and Avram (2015)

- The L1-acquisition process of DOM builds on the core semantic features of animacy and specificity (corpus study, Romanian and Spanish)

Guijarro-Fuentes (2011)

- Identifies animacy as the most influential factor in L2-acquisition of DOM (L1-English, acceptability judgement task)

Mayer and Sánchez (2021)

- Identify animacy as the most influential factor in L2-acquisition of DOM (L1-acquisition)

Parodi and Avram (2018)

- Animacy-based cases of DOM are acquired more easily, whereas cases of DOM conditioned by pragmatic factors are mastered less well (L1-English acquisition of DOM)

Simon (2022)

- L1-English speakers have higher omission rates of DOM compared to natives (CEDEL2 corpus study)

RQ1:

Do L1-German speakers apply DOM in their L2 Spanish differently compared to native speakers?

RQ2:

In what ways do the L2 speakers differ from L1 speakers in their use of DOM?

→ Idea: Directly compare DOM behaviour of L1 and L2 speakers

CEDEL2 (Corpus Escrito del Español como L2)

- Open-access learner corpus of mostly written but also spoken vernacular Spanish (Lozano 2022)
- data collection: 2020 -
- Contains an L1-Spanish reference corpus
- Metadata: linguistic background variables (e. g. proficiency, age) and task variables

L2 proficiency

Proficiency level	Placement test score	Corresponding score
Lower beginner	0-12	0-28%
Upper beginner	13-20	30-47%
Lower intermediate	21-28	49-65%
Upper intermediate	29-35	67-81%
Lower advanced	36-40	83-93%
Upper advanced	41-43	95-100%

Table: Proficiency measurement (Lozano 2022: Metadata)

CEDEL2: Corpus Escrito del Español L2 (version 2)

v2.0
Sept. 2020[i CEDEL2](#)[Q Search / Download](#)[User guide](#)[Statistics](#)[About](#)[Contact](#)

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What is CEDEL2?

CEDEL2 stands for Corpus Escrito del Español como L2 (L2 Spanish Written Corpus).

CEDEL2 is a large database containing the language produced by learners of Spanish as a second/foreign language (L2). This database is called a language 'corpus'.

This is the new version of CEDEL2 (version 2), which already incorporates the data from the earlier version (CEDEL2 version 1).

CEDEL2 currently amounts to a total of 4,399 participants and 1,105,936 words, which makes it one of the largest corpora of its kind.

CEDEL2 holds data from learners of Spanish with different L1 backgrounds (where 'L1' means the learners' mother tongue and 'L2' their foreign language):

- L1 English-L2 Spanish
- L1 German-L2 Spanish
- L1 Dutch-L2 Spanish
- L1 French-L2 Spanish
- L1 Portuguese-L2 Spanish
- L1 Italian-L2 Spanish



Figure: CEDEL2 homepage¹

¹Click here to go to CEDEL2 website

L1-Spanish	L2-Spanish
20 written texts from different speakers	82 written texts from different speakers
4.500 words	16.000 words
Task 14 (Charlie)	Task 14 (Charlie)

Table: Overview of the L1-Spanish and L2-Spanish corpora

Data collection: Task 14 (Chaplin video²)

- *Mira el siguiente video de Charles Chaplin (4 min.) Haz un resumen de la historia. Puedes ver el video más de una vez.*
- *'Watch the following Chaplin video clip (4 min.) and retell the story. Summarise the story. You can watch the video clip more than once.'*

²Click here to go to the Chaplin video

- Manual coding of sentences
- Selection of sentences containing nominal direct objects (exclusion of personal pronouns)

Variable/Factor	Levels
DOM	marked, unmarked
ANIMACY	animate, inanimate
SPECIFICITY*	specific, unspecific
GIVENNESS	given, new
CONTINUITY	continuous, discontinuous
TELICITY	telic, atelic
AFFECTEDNESS	unaffected, affected, highly affected

Table: Variables and manifestations annotated for L1 Spanish and L2 Spanish

L1 Spanish						
variable	level	frq #	frq %	level	frq #	frq %
DOM	marked	101	0.46	unmarked	118	0.54
ANIMACY	animate	113	0.52	inanimate	106	0.48
SPECIFICITY	specific	133	0.61	unspecific	86	0.39
GIVENNESS	given	94	0.43	new	125	0.57
CONTINUITY	continuous	17	0.08	discontinuous	202	0.92
TELICITY	telic	129	0.59	atelic	90	0.41
AFFECTEDNESS	affected, highly affected	191	0.87	unaffected	28	0.13
TOTAL		219				1.0

Figure: Overview of token numbers according to variables in the L1 Spanish corpus.

L2 Spanish						
variable	level	frq #	frq %	level	frq #	frq %
DOM	marked	217	0.25	unmarked	651	0.75
ANIMACY	animate	605	0.70	inanimate	263	0.30
SPECIFICITY	specific	550	0.63	unspecific	318	0.37
GIVENNESS	given	463	0.53	new	405	0.47
CONTINUITY	continuous	201	0.23	discontinuous	667	0.76
TELICITY	telic	574	0.66	atelic	294	0.34
AFFECTEDNESS	affected, highly affected	714	0.82	unaffected	154	0.18
TOTAL		868				1.0

Figure: Overview of token numbers according to variables in the L2 Spanish corpus.

Multifactorial Prediction and Deviation Analysis using Random Forests

- Developed by Deshors and Gries (2016)
- Approach in which two models are created using random forests
- Increasingly used in the investigation of varieties and L2 acquisition

What are random forests?

- Developed by Breimann (2001)
- Ensembles of decision trees, whereby randomly sampled data is used to arrive at an aggregated estimate of a particular outcome's probability through random permutation tests (Breimann 2001: 113; Strobl et al. 2009: 325)
- Due to their non-parametric nature, random forests are much less affected by data sparsity, collinearity, and overfitting (Breimann 2001: 113; Gries 2022: 618, Tagliamonte and Baayen 2012)

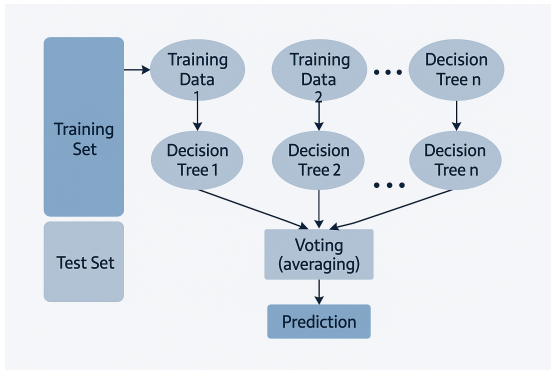


Figure: Infographic random forests

Limitations of random forests

- Difficulty of visualising and interpreting the final result (recommendation: use of representative trees instead of conditional inference trees)
- Variable importance measure (e.g. through permutation) is necessary to evaluate the power of a predictor

Step 1:

- Fitting a first random forest model (RF1) on the reference speakers' (L1 speakers) choice of using DOM (Gries and Wulff 2021)

Step 2:

- Checking the goodness-of-fit of RF1 (via confusion matrices and variable importance)

Step 3:

- Applying the RF1 model to the target speaker group (L2-Spanish) to compute for every target speaker choice what a reference speaker would have done (Gries and Deshors 2020); adding a new factor/variable for `REFERENCELIKENESS`

Step 4:

- Fitting a second random forest (RF2) with `REFERENCELIKENESS` as response variable and `DOM` as a predictor to see when L2 speakers behaved target-like (Gries 2022: 257); checking the goodness-of-fit of the model (cf. Step 2)

- Fitting a first random forest model (RF1) on the reference speakers' choice of using DOM (Gries and Wulff 2021)

	unmarked	marked
unmarked	106	12
marked	0	101

Table: Confusion matrix of the predictions of the first random forest (RF1) for DOM, C-score = 94.5%; OOB = 5.48%

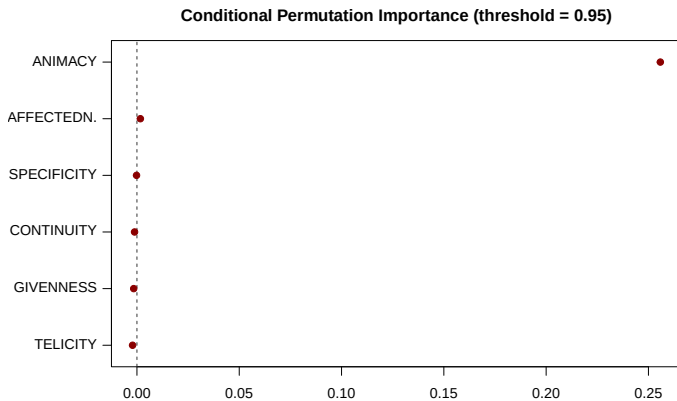


Figure: Variable importance of RF1

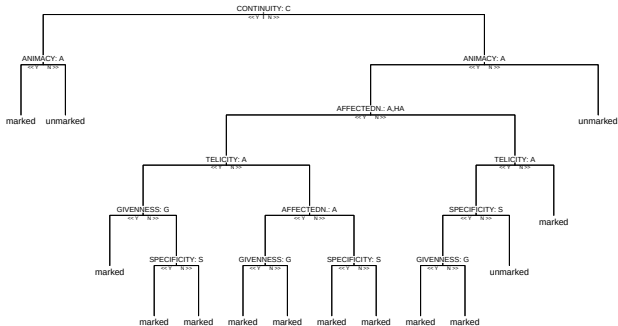
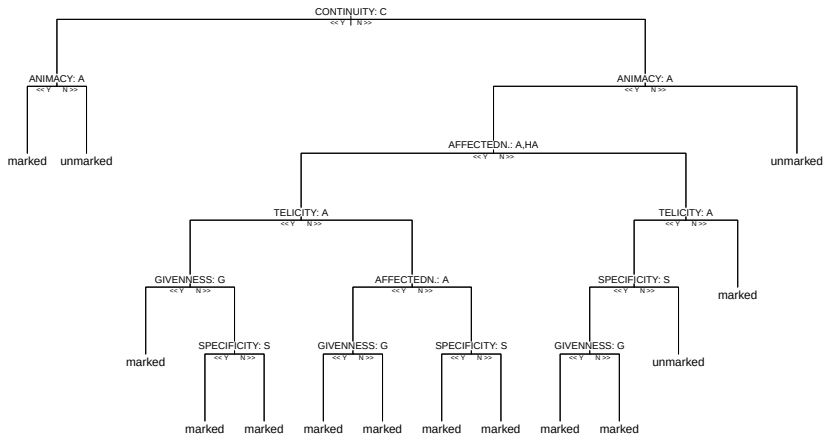


Figure: Representative tree of the random forest RF1



- With REFERENCELIKENESS as response variable and DOM as a predictor to see when L2-speakers behaved target-like

	MATCH	NO_MATCH
MATCH	485	0
NO_MATCH	3	380

Table: Confusion matrix of the predictions of the second random forest (RF1) for REFERENCELIKENESS, C-score = 99.4%; OOB = 0.34%

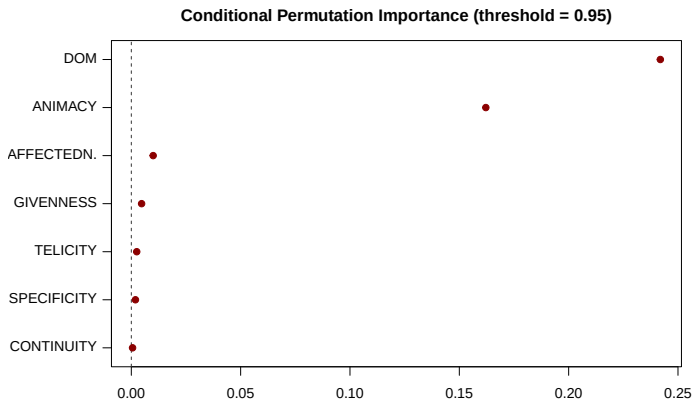


Figure: Variable importance of RF2

6.1 Representative tree of RF2

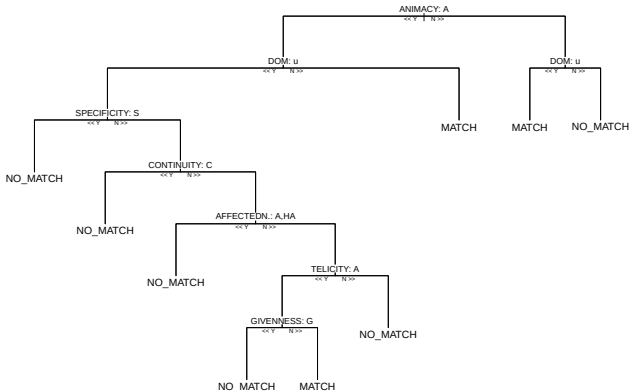
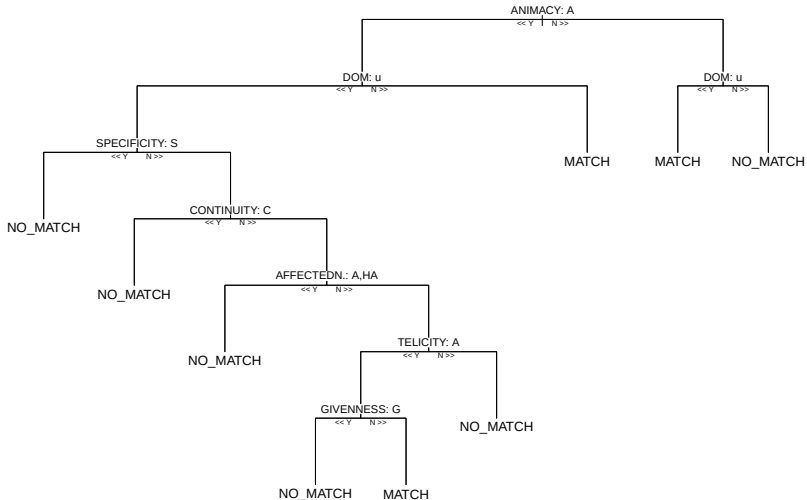


Figure: Representative tree of the random forest RF2



RQ1:

Do L1-German speakers apply DOM in their L2 Spanish differently compared to native speakers?

RQ2:

RQ2 In what ways do the L2 speakers differ from L1 speakers in their use of DOM?

Findings based on RF2

- Learners performed completely referencelike with regard to their a-marked direct objects
- Learners chose to leave a considerable number of animate direct objects unmarked that would have been marked by natives
- ANIMACY as the key predictor of REFERENCELIKENESS since it is able to account for the targetlike and non-targetlike behaviour of the learners

Implications on processing of DOM

- Learners seem to have a general understanding of DOM, but have not acquired it to the point of automation

Possible explanations

- Potential negative L1-transfer or the lack of positive transfer from German, which lacks a DOM system (Schmitz 2006: 245)
- Considering that the omission of the marker only leads to misunderstandings in very few cases, attention to DOM can remain low without penalty in most communicative contexts

- This analysis sheds new light on the proportional influence and interactions of the semantic and pragmatic factors which are assumed to influence the use of DOM
- The results indicated that animacy is by far the most influential predictor, with the others only displaying marginal to no significant importance
- Limitations: could be due to the small L1 dataset and the specific task environment

¡Gracias!

¿Preguntas sobre DOM, los datos, el análisis?

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Studies that have used CEDEL2

If you have used CEDEL2 as a source of data in your publication / dissertation and want to see it advertised here, please send the details to [Cristóbal Lozano](#)

Papers/chapters

1. **Quesada, T., & Lozano, C. (2025).** Bidirectionality in bilingualism? Asymmetry in L1 Spanish-L2 English vs. L1 English-L2 Spanish bilinguals. *Linguistics*, 103898.
2. **Kocher, A. (2025).** The subject realization in L2 Spanish by German L1 speakers: A corpus study. *Isogloss. Open Journal of Romance Linguistics*, Article 2.
3. **Adolf, P. (2024).** Differential object marking in the L2 Spanish of L1 German speakers: A corpus-based analysis. In: Grünke, J., Pešková, A., Gabr (eds) *Spanish as a second and third language*. Romanistik, vol 42. Frank & Timme, Berlin.

Figure: Previous studies conducted with CEDEL2 data

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